

Algebra 1: Polynomials

Classwork Level

Name: _____

Date: _____

Question 1: Simplify the expression $x^2 + 3x + 2$ by factoring.

A) $(x + 1)(x + 2)$

B) $(x + 2)(x - 1)$

C) $(x - 1)(x - 2)$

D) $(x - 2)(x - 1)$

E) $(x + 3)(x - 1)$

Question 2: If $p(x) = x^2 + 4x + 4$, what is $p(-2)$?

A) 0

B) -4

C) 4

D) 8

E) 12

Question 3: Given $f(x) = x^3 - 6x^2 + 11x - 6$, factor $f(x)$ completely and identify the zeros of the function.

Question 4: Which polynomial has a greater degree, $3x^2 + 2x - 1$ or $2x^3 - x^2 + x$?

A) $3x^2 + 2x - 1$

B) $2x^3 - x^2 + x$

C) Both have the same degree

D) Cannot be determined

E) Neither

Question 5: Simplify $(2x^2 + 3x - 1) + (x^2 - 2x - 1)$.

A) $3x^2 + x - 2$

B) $3x^2 + x - 1$

C) $x^2 + x - 2$

- D) $x^2 - x - 2$
- E) $3x^2 + 5x - 2$

Question 6: If $p(x) = 2x^3 - x^2 - 6x + 3$, what is $p(2)$?

- A) 1
- B) 3
- C) 5
- D) 7
- E) 11

Question 7: Find all the zeros of the function $f(x) = 2x^3 + 3x^2 - 8x - 12$, showing all work and justifying your answers.

Question 8: Given $f(x) = x^4 - 5x^3 + 3x^2 + x - 1$, what is the degree of the polynomial?

- A) 3
- B) 4
- C) 5
- D) 6
- E) 7

Question 9: If $g(x) = 3x^2 - 2x - 5$, what is the value of $g(0)$?

- A) -2
- B) -5
- C) 0
- D) 2
- E) 5

Question 10: Prove that for any polynomial $p(x)$, if $p(a) = 0$, then $(x - a)$ is a factor of $p(x)$. Provide a step-by-step solution and explain the significance of this result in polynomial factorization.

Chef's Tips (Hints):

- To factor polynomials, look for common factors first.
- When simplifying expressions, combine like terms to make the process easier.
- To find zeros of a function, set the function equal to zero and solve for x .